

ASEN 3502 Computational Methods

Homework 4

See Gradescope for Due Date

This homework is an individual pre-lab for Lab 3. You will implement and verify Euler's method on a test problem with a known exact solution. You will use this integrator to propagate orbits in Lab 3, and you will reuse your test function to test RK4.

You will **submit one pdf** with code and plots combined, generated from a **single source file**. Define `euler`, `test_dyn`, and `test_ode` as local functions at the end of that file. You can use any language; the pdf can be created with `publish` or Live Script export in Matlab, `Weave.jl` in Julia, or `jupyter nbconvert --to pdf` on a Jupyter notebook in Python.

Starter code is provided in `hw4_starter.m`

Question 1 (Implement Euler's method)

- (a) Show the explicit Euler update equation for $\{\dot{x}\} = \{f\}(t, \{x\})$ with step size h .
- (b) Implement an integrator that uses this update in a function with the following interface, which follows the argument order of `ode45`:

```
function [t, x] = euler(dynamics, tspan, x0, h)
```

Here `dynamics(t, x)` returns the state derivative, `tspan = [t0 tf]` gives the requested interval, `x0` is the initial state, and `h` is a positive step size. Like `ode45`, return a column vector of times and a state history with one state per row, so that `x(i, :)` is the state at time `t(i)`. Your function should support any state dimension compatible with the dynamics.

- (c) The system of first-order equations

$$\{f\}(t, \{x\}) = \begin{Bmatrix} x_2 \\ -x_1 \end{Bmatrix}, \quad \{x\}(0) = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$$

has the exact solution $x_1(t) = \cos t$, $x_2(t) = -\sin t$. Use your `euler` function to solve this system with $h = 0.1$ on $0 \leq t \leq 10$. Plot the numerical $x_1(t)$ together with the exact solution and describe how they differ.

Question 2 (Verify Euler's method)

Consider

$$\ddot{y} = a^2 y, \quad y(0) = 1, \quad \dot{y}(0) = a, \quad a = 1, \quad 0 \leq t \leq 1.$$

- (a) Derive the exact solution.
- (b) Express this equation as a two-state first-order system and implement it in `test_dyn(t, x)`.
- (c) Create a function `test_ode(integrator, h, atol)`. It should call the specified integrator on this test problem and compare the first state component with the exact solution at every returned time. Use `assert` to raise an error unless the maximum absolute error is strictly less than `atol`.

(d) Run `test_ode(@euler, 0.001, 2e-3)` and show that it completes without an error.

Question 3 (Error trend)

For the same test problem, use

$$h = 1/N, \quad N = \{10, 20, 40, 80, 160, 320, 640\}.$$

For each step size, calculate the maximum absolute error

$$E_{\max}(h) = \max_i |y(t_i) - x_{1,i}|$$

over all returned times, where $x_{1,i}$ is the first component of the numerical state at t_i .

- (a) Plot $E_{\max}$ against h on logarithmic axes.
- (b) Estimate the slope and relate it to the expected global order of Euler's method.
- (c) Based on your results, estimate the step size required to reach $E_{\max} = 10^{-6}$.